Classical Field Theory — Exercise Sheet 9

Gauge identities and source conditions

Solutions

Use $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. Unless a weaker class is explicitly discussed, fields and coefficients are smooth. Compact support is stated separately in each problem.

Exercise 9.1 (The adjoint differentiates coefficients too). Let a smooth first-order density $L(x,\phi,\partial\phi)$ have Euler expressions E_a on an open coordinate region $U \subset \mathbb{R}^{1+d}$. Vary its action on a compact region $D \subset U$, with every parameter supported in the interior of D. For smooth parameters ϵ^α, consider the finite-order variation

$$\delta_\epsilon\phi^a = \sum_{|I|\le r} R^{aI}_\alpha(x, j^s\phi)\partial_I\epsilon^\alpha.$$

Here I is a derivative multi-index, each multi-index occurs once in the sum, and $j^s\phi$ denotes the field and all its derivatives through finite order s. The coefficients are smooth functions of these arguments. Assume the action is invariant under these variations for every compact parameter and every smooth field, without imposing the field equations.

(a) Derive the identity satisfied by the Euler expressions, including the variable coefficients. For the one-dimensional test operator $R\epsilon = (1+x^2)\epsilon_x + x\epsilon$, display R^*E explicitly. This test asks for its adjoint, not for a claim that every density has that symmetry.

(b) For Maxwell's transformation $\delta\phi_{\mathrm{em}} = -\epsilon_t$, $\delta\mathbf{A} = \nabla\epsilon$, derive the identity for the scalar and vector Euler expressions. With $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}$, include a prescribed source $(\rho,\mathbf{J})$ and identify its necessary conservation condition. Explain the distributional interpretation for only C^2 potentials.

Solution. The parameter and all its derivatives have compact support, so the field variation does too. Lecture 1's compact-variation formula gives

$$0 = \delta_\epsilon S = \int \sum_{a,I} E_a R^{aI}_\alpha \partial_I\epsilon^\alpha \mathrm{vol} = \int \sum_{a,I} (-1)^{|I|}\partial_I(E_a R^{aI}_\alpha)\epsilon^\alpha \mathrm{vol}.$$

Every boundary term vanishes by compact support. The fundamental lemma gives one identity for each α. These derivatives act after the coefficient has been evaluated on the field, including its implicit coordinate dependence through the field derivatives. For the test operator, ordinary integration by parts gives

$$R^*E = -\partial_x((1+x^2)E) + xE = -(1+x^2)E_x - xE.$$

Dropping the derivative of $1+x^2$ would leave the incorrect coefficient $+xE$ instead of $-xE$.

For Maxwell the compact variation pairs as $\int[-E_\phi\epsilon_t + E_A\cdot\nabla\epsilon]\mathrm{vol}$. Its adjoint is $\partial_t E_\phi - \operatorname{div}E_A$. Lecture 9's density gives

$$E_\phi = \mathcal{M}_0 - \rho, \qquad E_A = \boldsymbol{\mathcal{M}} + \mathbf{J}.$$

The source-free identity follows directly from commuting derivatives: $\partial_t\mathcal{M}_0 - \operatorname{div}\boldsymbol{\mathcal{M}} = 0$. Consequently

$$\partial_t E_\phi - \operatorname{div}E_A = -\rho_t - \operatorname{div}\mathbf{J}.$$

This vanishes identically exactly for a conserved source. For a C^2 potential the Euler expressions are continuous but need not have classical first derivatives. Pair the identity with compact smooth tests and move derivatives onto those tests. Distributional mixed derivatives commute, which proves the same identity without assuming third derivatives of the potential.

Exercise 9.2 (Polarization and magnetization). Let $P, M \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R}^3)$ and define

$$\rho = -\operatorname{div} P, \qquad \mathbf{J} = \partial_t P + \operatorname{curl} M.$$

Prove conservation, zero total charge on each slice and $\int y\rho(t,y)\mathrm{d}^3y = \int P(t,y)\mathrm{d}^3y$. For any smooth potential, rewrite the integrated source coupling $\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}$ in terms of P, M and the fields, displaying why all discarded terms vanish. Does this construction assert that the potential of this source is unique?

Solution. Differentiation and commutation of spatial derivatives give

$$\rho_t + \operatorname{div}\mathbf{J} = -\operatorname{div} P_t + \operatorname{div} P_t + \operatorname{div}\operatorname{curl} M = 0.$$

Compact spatial support removes every face integral, so $\int \rho = 0$. For its first moment, integrating each coordinate derivative gives

$$\int y_i\rho(t,y)\mathrm{d}^3y = -\int y_i\partial_{y_j}P_j(t,y)\mathrm{d}^3y = \int P_i(t,y)\mathrm{d}^3y.$$

Substitution in the coupling and integration by parts give

$$\begin{aligned}\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol} &= \int[(\operatorname{div} P)\phi_{\mathrm{em}} + P_t\cdot\mathbf{A} + (\operatorname{curl} M)\cdot\mathbf{A}]\mathrm{vol}\\ &= \int[-P\cdot\nabla\phi_{\mathrm{em}} - P\cdot\partial_t\mathbf{A} + M\cdot\operatorname{curl}\mathbf{A}]\mathrm{vol}\\ &= \int(P\cdot\mathbf{E} + M\cdot\mathbf{B})\mathrm{vol}.\end{aligned}$$

The spatial face terms contain P or M; the time-face term contains P. All vanish outside a fixed compact set, regardless of the smooth potential's behaviour at infinity. The curl sign also follows from $\operatorname{div}(M\times\mathbf{A}) = \mathbf{A}\cdot\operatorname{curl} M - M\cdot\operatorname{curl}\mathbf{A}$. The final expression depends only on the fields and is gauge invariant. It supplies a family of compatible sources, not a uniqueness theorem for potentials: adding $\mathrm{d}\chi$ still leaves their fields unchanged.

Exercise 9.3 (Vacuum duality with the Lorentzian star). For a vacuum Maxwell field, put $F' = \star F_{\mathrm{em}}$, using exactly the Hodge table of Lecture 9 and $\star^2 = -1$ on two-forms. Find the transformed electric and magnetic vectors, and verify all four vacuum component equations. Determine the effect of two and four successive transformations. Check the energy and Poynting densities for the constant vacuum fields $\mathbf{E} = (1,0,2)$, $\mathbf{B} = (0,3,0)$ before and after the map. These are densities, not finite total energies on $\mathbb{R}^3$. Does the same transformation preserve Maxwell's equations with a prescribed nonzero electric current three-form $\mathcal{J}$ and no magnetic source?

Solution. The Hodge table gives $\star(\mathrm{d}x^i\wedge\mathrm{d}t)$ equal to the corresponding spatial magnetic wedge, and the star of that wedge is $\mathrm{d}t\wedge\mathrm{d}x^i = -\mathrm{d}x^i\wedge\mathrm{d}t$. Thus $E'_i = -B_i$ and $B'_i = E_i$. The transformed divergences vanish because $\operatorname{div}\mathbf{E}' = -\operatorname{div}\mathbf{B} = 0$ and $\operatorname{div}\mathbf{B}' = \operatorname{div}\mathbf{E} = 0$. The two evolution equations become

$$\partial_t\mathbf{B}' + \operatorname{curl}\mathbf{E}' = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B} = 0, \qquad \operatorname{curl}\mathbf{B}' - \partial_t\mathbf{E}' = \operatorname{curl}\mathbf{E} + \partial_t\mathbf{B} = 0.$$

Applying the map twice changes the sign of both fields and four times restores them.

In the example the transformed fields are $(0,-3,0)$ and $(1,0,2)$. Both squared-magnitude sums are 14, and direct cross products give

$$\frac{1}{2}(|\mathbf{E}|^2+|\mathbf{B}|^2)=7, \qquad \mathbf{E}\times\mathbf{B}=(-6,0,3)=\mathbf{E}'\times\mathbf{B}'.$$

For a prescribed electric source, $\mathrm{d}F_{\mathrm{em}}=0$ but $\mathrm{d}{\star}F_{\mathrm{em}}=\mathcal{J}$. Hence $\mathrm{d}F'=\mathcal{J}$ and $\mathrm{d}{\star}F'=-\mathrm{d}F_{\mathrm{em}}=0$. These are not the original equations with the same nonzero electric source; the transformed Bianchi equation contains a magnetic-source term.

Exercise 9.4 (Conservation does not remove a time-face term). Let $D=[t_0,t_1]\times\mathbb{T}^3$, $t_1>t_0$, and use the Maxwell density $L=\frac{1}{2}(|\mathbf{E}|^2-|\mathbf{B}|^2)-\rho\phi_{\mathrm{em}}+\mathbf{J}\cdot\mathbf{A}$. Take smooth periodic fields and a prescribed smooth conserved source. Allow a smooth periodic gauge function $\chi(t,x)$ with nonzero time-face values.

(a) Find the exact action change, retaining the time faces and explaining the cancellation on opposite spatial faces.

(b) Set $\rho=\sin x^1$, $\mathbf{J}=0$, $\phi_{\mathrm{em}}=\sin x^1$, $\mathbf{A}=0$, and $\chi=\gamma t\sin x^1$, where real $\gamma\neq 0$. Verify Maxwell's equations before and after the transformation and compute the nonzero action change. Why is this consistent with Lecture 9's compact-parameter gauge invariance?

Solution. A gauge change leaves the two field strengths fixed, so only the source coupling changes. Its exact difference is

$$\Delta S=\int_D(\rho\chi_t+\mathbf{J}\cdot\nabla\chi)\mathrm{vol}=\int_D[\partial_t(\rho\chi)+\mathrm{div}(\chi\mathbf{J})]\mathrm{vol},$$

where conservation removed the remaining bulk term. Periodicity cancels opposite spatial faces, leaving

$$\Delta S=\left[\int_{\mathbb{T}^3}\rho(t,x)\chi(t,x)\mathrm{d}^3x\right]_{t_0}^{t_1}.$$

In the example, the static electric field is $(-\cos x^1,0,0)$ and its divergence is $\sin x^1=\rho$. The magnetic field and spatial current are zero; all remaining Maxwell equations hold. The source is conserved and neutral. After the gauge change the potentials are

$$\phi'_{\mathrm{em}}=(1-\gamma)\sin x^1, \qquad \mathbf{A}'=(\gamma t\cos x^1,0,0).$$

Their electric contributions add to the same $-\cos x^1$, and their curl is zero. Thus the same Maxwell equations hold. Finally,

$$\Delta S=\gamma(t_1-t_0)\int_{\mathbb{T}^3}\sin^2 x^1\mathrm{d}^3x=4\pi^3\gamma(t_1-t_0)\neq 0.$$

Lecture 9 imposed compact support of the gauge parameter away from every boundary. The present parameter has different nonzero face data, so that hypothesis is absent. Local field invariance and source conservation alone do not remove this boundary charge pairing.